

BROWNSVILLE, TX, USA

WMO: 722500

Lat: 25.9167N

Lon: 97.4192W

Elev: 26

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.6	42.4	24.8	18.9	50.9	30.0	24.1	51.0	29.2	77.3	26.9	76.9	10.9	330	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.7	96.2	78.4	94.9	78.3	93.7	78.2	81.5	88.3	80.8	87.8	80.3	87.4	15.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	155.9	83.6	79.2	151.7	83.3	78.5	147.9	83.0	45.2	88.5	44.4	87.8	43.7	87.6	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	24.1	21.0	DB	33.9	99.7	3.6	2.3	31.3	101.4	29.1	102.7	27.1	104.0	24.5	105.7	(4)
(5)			WB	30.9	82.7	3.4	1.2	28.5	83.6	26.5	84.3	24.6	85.0	22.2	85.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.1	61.9	65.4	69.8	75.4	80.9	84.4	85.1	85.7	82.3	76.8	69.7	63.3	(6)	
	DBStd	10.47	8.88	8.75	7.21	5.98	4.13	2.46	2.14	2.33	3.50	5.97	7.79	8.87	(7)	
	HDD50	38	17	8	1	0	0	0	0	0	0	0	2	10	(8)	
	HDD65	499	164	96	40	7	0	0	0	0	0	5	45	142	(9)	
	CDD50	9194	385	439	614	762	957	1033	1088	1105	968	830	592	421	(10)	
	CDD65	4181	67	107	188	319	492	583	623	640	518	370	186	88	(11)	
	CDH74	42220	298	521	995	2398	4949	6974	7703	7902	5440	3320	1265	455	(12)	
	CDH80	17484	26	82	201	731	1866	3226	3692	3889	2284	1153	281	53	(13)	
(14) Wind	WSAvg	10.3	10.1	11.3	11.7	12.5	12.5	11.1	10.9	9.3	7.5	8.4	9.0	9.5	(14)	
(15) Precipitation	PrecAvg	26.4	1.3	1.1	0.9	1.5	2.6	2.6	1.7	2.5	5.8	3.3	1.5	1.3	(15)	
	PrecMax	47.0	4.8	4.7	5.9	6.6	9.7	8.9	9.4	9.6	20.2	13.7	5.6	5.6	(16)	
	PrecMin	15.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0	0.0	(17)	
	PrecStd	6.6	1.1	1.1	1.2	1.6	2.2	2.2	1.8	2.0	4.1	3.0	1.3	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.7	86.0	88.3	92.4	94.4	96.7	97.2	98.0	96.6	92.4	88.0	84.0	(19)	
		MCWB	69.3	69.4	72.7	73.7	78.0	78.9	78.4	78.9	77.5	77.8	74.7	72.5	(20)	
	2%	DB	80.0	82.4	84.3	89.1	92.1	94.4	95.4	96.3	93.8	90.3	85.3	81.4	(21)	
		MCWB	69.0	71.4	72.0	73.9	77.4	78.6	78.3	78.6	77.8	77.2	73.7	71.8	(22)	
	5%	DB	77.5	80.0	82.1	86.8	90.3	93.1	94.2	95.1	91.9	88.4	83.4	79.2	(23)	
		MCWB	67.6	70.2	71.0	74.1	76.8	78.3	78.2	78.4	77.9	76.4	72.7	70.5	(24)	
	10%	DB	74.7	77.2	79.7	84.4	88.5	91.8	92.8	93.7	90.2	86.4	81.1	76.3	(25)	
		MCWB	66.5	68.3	69.8	73.1	76.4	78.1	78.2	78.4	77.7	75.3	71.7	69.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.0	74.3	75.5	78.6	81.2	82.8	81.6	82.4	81.8	81.4	78.2	75.2	(27)	
		MCDB	78.2	80.4	83.1	86.2	88.1	91.1	89.1	89.3	87.5	87.5	83.2	80.0	(28)	
	2%	WB	71.4	73.0	74.2	77.1	80.0	81.4	80.6	81.3	80.6	80.0	76.6	73.7	(29)	
		MCDB	76.1	78.4	80.7	84.6	87.0	87.7	88.2	89.0	86.9	85.7	81.3	78.2	(30)	
	5%	WB	70.2	72.0	73.1	76.0	79.0	80.5	80.1	80.4	80.1	78.8	75.3	72.6	(31)	
		MCDB	74.8	76.8	78.8	82.9	85.9	87.1	87.8	88.3	86.6	84.4	80.0	76.8	(32)	
	10%	WB	68.9	70.8	72.1	75.1	78.1	79.9	79.5	79.9	79.4	77.8	73.9	71.1	(33)	
		MCDB	73.2	75.2	76.9	81.3	84.6	86.8	87.3	88.0	86.2	83.4	78.9	75.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	17.6	16.7	16.4	15.1	15.5	15.8	16.7	15.9	17.1	17.8	18.0	(35)	
		MCDBR	19.2	17.8	17.3	17.5	16.1	16.8	17.5	18.0	17.4	17.4	17.8	18.3	(36)	
	5% WB	MCWBR	10.1	8.0	7.1	6.4	4.5	4.2	3.9	3.9	5.2	6.6	8.2	10.1	(37)	
		MCWBR	16.3	15.5	14.9	15.3	13.6	14.4	15.5	16.2	14.6	14.4	15.1	15.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.368	0.381	0.402	0.450	0.463	0.434	0.453	0.439	0.441	0.393	0.379	0.371	(40)	
		taud	2.464	2.418	2.355	2.223	2.218	2.364	2.320	2.367	2.330	2.461	2.458	2.456	(41)	
	Ebn at Noon		280	285	284	271	266	272	267	271	268	277	274	273	(42)	
		Edh at Noon	31	35	39	46	46	39	41	39	40	33	31	30	(43)	
(44) All-Sky Solar Radiation	RadAvg		990	1195	1471	1752	1949	2101	2084	2002	1664	1458	1120	898	(44)	
	RadStd		107	168	148	140	112	74	139	81	113	128	113	119	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+268	+266	(46)	
(47)	Regional (12 neighbors)	N/A	N/A	N/A	+0.56	N/A	N/A	N/A	N/A	N/A	+227	(47)	

Nomenclature: See separate page